

Design FMEA (qualitative)

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Qualitative version based on simulation risk signals (not a full process/supplier FMEA). | **Date**: 2026-08-25

Failure mode	Cause	Simulation signal	S	O	Design-side mitigation (implemented)
Negative electrode lithium plating (fast charge)	anode polarization at 4C, limited negative headroom	min anode potential 0.0247 V (>0 → no plating signal; V1/V2 showed negative/edge signals)	High	Low	nano particles, high-transport electrolyte (t ₀ =0.50 est.), thinner electrodes, separator porosity 0.55
Thermal runaway risk (temperature rise beyond limit)	4C charge heat generation vs h=10 cooling	4C T_max 331.98 K vs 333.15 (margin 1.2 K; V3/V5 exceeded)	High	Low	transport upgrade + larger pouch cooling area 0.0075 m² (V5 control shows +2.8 K without it)
Electrolyte oxidative decomposition	voltage window vs HOMO/IE-EA	Not computed (no molecular funnel; real_compute=false)	Medium	Medium	upper cutoff 4.2 V conservative; LNMO 4.7 V system rejected (lower ED path)
Insufficient capacity / energy	loading too low, rate losses	1C 3.45 Ah, ED 473.0 Wh/kg vs targets	Medium	Low	thickness/Pareto scan V1–V5; final balances ED vs 5C vs T_max
Overcharge thermal runaway	charge beyond 4.2 V	0.5C to 4.7 V: T_max 300.57 K, triggered=false	High	Low	BMS cutoff required (beyond cell design)

Conclusion: Highest-risk item = 4C fast-charge thermal (margin 1.2 K); mitigation implemented via transport + cooling-area levers; plating risk mitigated. Complete FMEA including process/supplier failures: N/A (beyond pure-simulation boundary).